FWD and AFT Avionic Shelves – General data

1. General

Avionics components are installed on equipment racks located in the forward fuselage and in the aft empennage structure.

2. Description

The forward avionics equipment racks are entirely contained within the metallic section of the forward fuselage of the aircraft with the exception of the top shelf which has been extended into the nose baggage compartment to allow for mounting of the Apex MAU hardware. To provide protection from HIRF effects, the metallic structure has been extended into the baggage compartment to fully enclose the Apex avionics equipment within a metallic envelope. Metallic covers are provided to encase the entire avionics bay.

The aft avionics equipment racks are entirely contained within the metallic section of the aft empennage structure of the aircraft.

3. Adjustment/Test

A. Avionics Equipment Rack Test

Measure the resistance between the equipment rack and the airframe using a milliohm meter, resistance to be less than 2.5 milliohms. Probe measurements to be taken on the equipment rack mounting skin surface (as close as possible to the bonding stud bracket) and the corresponding airframe bonding stud.

B. Avionics Equipment Rack Adjustment

Rework bonding jumper attachments and/or brackets as required in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data to achieve bonding requirement.

4. Removal and Installation

A. Forward Avionics Equipment Racks

1) Removal of Equipment Racks

a) Gain access to the forward avionics racks through the forward baggage compartment.

b) Remove the top and middle covers attached to bulkhead at stations 37.00 & 44.00.

c) Disconnect and remove electrical components as required to allow removal of the equipment rack(s) (refer to associated individual component maintenance practices and 20-70-01, Handling of Static Sensitive Electronic Components and Devices – General Data).

d) Remove the bonding jumpers attached to the equipment rack(s).

e) Remove equipment rack fasteners and remove the equipment rack(s) from the avionics compartment.

2) Installation of Equipment Racks

a) Set equipment rack(s) in position and install fasteners.

b) Install electrical components and associated wiring (refer to individual component maintenance practices and 20-70-01, Handling of Static Sensitive Electronic Components and Devices – General Data).

c) Attach disconnected bonding jumpers to equipment rack(s).

d) Verify electrical bonding of equipment rack(s) to airframe is maintained. Refer to Para 3.A.

e) Install the top and middle covers attached to bulkhead at stations 37.00 and 44.00. Ensure the covers are mounted securely.

WARNING **ACCESS COVERS ARE REQUIRED TO MAINTAIN PROTECTION FEATURES THAT ARE ESSENTIAL TO MINIMIZING THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF).**

f) Close forward baggage compartment door.

B. Aft Avionics Equipment Racks

1) Removal of Equipment Racks

a) Gain access to the aft avionics racks through the aft baggage compartment.

b) Disconnect and remove electrical components as required to allow removal of the equipment rack(s) (refer to associated individual component maintenance practices and 20-70-01, Handling of Static Sensitive Electronic Components and Devices – General Data).

c) Remove the bonding jumpers attached to the equipment rack(s).

d) Remove equipment rack fasteners and remove the equipment rack(s) from the avionics compartment.

2) Installation of Equipment Racks

a) Set equipment rack(s) in position and install fasteners.

b) Install electrical components and associated wiring (refer to individual component maintenance practices and 20-70-01, Handling of Static Sensitive Electronic Components and Devices – General Data).

c) Attach disconnected bonding jumpers to equipment rack(s).

d) Verify electrical bonding of equipment rack(s) to airframe is maintained. Measure the resistance between the equipment rack and the airframe using a milliohm meter, resistance to be less than 2.5 milliohms. Probe measurements to be taken on the equipment rack mounting skin surface (as close as possible to the bonding stud bracket) and the corresponding airframe bonding stud. Rework bonding jumper attachments and/or brackets as required in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data to achieve bonding requirement.

e) Close aft baggage compartment door.